1) a)

$\frac{x-4}{2} = \frac{y-1}{3} = \frac{z+2}{8}$

1) b)

$\frac{x-2}{9} = \frac{y+8}{-1} = \frac{z+5}{-2}$

1) c)

$\frac{x+3}{2} = \frac{y+5}{-1} = \frac{z-6}{-5}$ or $\frac{x+3}{2} = \frac{y+5}{-1} = \frac{6-z}{5}$

1) d)

$\frac{x-1}{-1} = \frac{y+1}{3} = \frac{z-0}{10}$ or $1 - x = \frac{y+1}{3} = \frac{z}{10}$

1) e)

$\frac{x-0}{-1} = \frac{y-8}{-5} = \frac{z+3}{9}$ or $-x = \frac{8-y}{5} = \frac{z+3}{9}$

1) f)

$\frac{x-\frac{1}{2}}{5} = \frac{y+3}{-2} = \frac{z+\frac{1}{3}}{-10}$ or $\frac{2x-1}{10} = \frac{y+3}{-2} = \frac{3z+1}{-30}$

1) g)

$\frac{x+\frac{1}{4}}{10} = \frac{y-\frac{1}{2}}{-4} = \frac{z-\frac{2}{3}}{-3}$ or $\frac{4x+1}{40} = \frac{1-2y}{8} = \frac{2-3z}{9}$

1) h)

$\frac{x-2}{3} = \frac{y+5}{4}$, $z = -7$

1) i)

$\frac{x+7}{8} = \frac{z-12}{-3}$, $y = 3$ or $\frac{x+7}{8} = \frac{12-z}{3}$, $y = 3$

1) j)

$x = -4$, $\frac{y-3}{-5} = \frac{z-3}{6}$ or $x = -4$, $\frac{3-y}{5} = \frac{z-3}{6}$

1) k)

$y = 3$, $z = -7$

1) l)

$\frac{x-\frac{3}{4}}{-16} = \frac{z-\frac{9}{4}}{3}$, $y = -\frac{5}{3}$ or $\frac{3-4x}{64} = \frac{4z-9}{12}$, $y = -\frac{5}{3}$

2)

When $z = 6$:

$\frac{x-1}{3} = \frac{y+2}{4} = \frac{6-5}{2} = \frac{1}{2}$

$\frac{x-1}{3} = \frac{1}{2} \Rightarrow x = \frac{5}{2}$

$\frac{y+2}{4} = \frac{1}{2} \Rightarrow y = 0$

So the point has coordinates $\left(\frac{5}{2}, 0, 6\right)$

3)

When $y = 8$:

$\frac{x+5}{6} = \frac{z+8}{-3} = \frac{2-8}{10} = -\frac{3}{5}$

$\frac{x+5}{6} = -\frac{3}{5} \Rightarrow x = -\frac{43}{5}$

$\frac{z+8}{-3} = -\frac{3}{5} \Rightarrow z = -\frac{31}{5}$

So the point has coordinates $\left(-\frac{43}{5}, 8, -\frac{31}{5}\right)$

4)

When $x = -5$:

$\frac{-5-3}{-2} = 4$

When $y = 20$:

$\frac{20-4}{4} = 4$

When $z = 6$:

$\frac{6+6}{3} = 4$

The values are consistent, so $(-5, 20, 6)$ lies on the line.

5)

When $x = 2$:

$\frac{2-9}{3} = -\frac{7}{3}$

When $y = -8$:

$\frac{-8-13}{9} = -\frac{21}{9} = -\frac{7}{3}$

The values are consistent, so $(2, -8, 3)$ lies on the line.

6)

When $x = 15$:

$\frac{15+2}{8} = \frac{17}{8}$

When $y = 11$:

$\frac{11+57}{32} = \frac{68}{32} = \frac{17}{8}$

When $z = -10$:

$\frac{-10+18}{4} = \frac{8}{4} = 2$

The values are not consistent, so $(15, 11, -10)$ does not lie on the line.

7) a)

$\begin{pmatrix} 2 \\ 3 \\ 6 \end{pmatrix} - \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} = \begin{pmatrix} 2 \\ 3 \\ 6 \end{pmatrix}$

$\frac{x}{2} = \frac{y}{3} = \frac{z}{6}$

7) b)

$\begin{pmatrix} 3 \\ 5 \\ -7 \end{pmatrix} - \begin{pmatrix} 1 \\ 2 \\ -4 \end{pmatrix} = \begin{pmatrix} 2 \\ 3 \\ -3 \end{pmatrix}$

$\frac{x-1}{2} = \frac{y-2}{3} = \frac{z+4}{-3}$

7) c)

$\begin{pmatrix} 5 \\ 1 \\ 0 \end{pmatrix} - \begin{pmatrix} 4 \\ 0 \\ -8 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \\ 8 \end{pmatrix}$

$\frac{x-4}{1} = \frac{y}{1} = \frac{z+8}{8}$ or $x - 4 = y = \frac{z+8}{8}$

7) d)

$$\begin{pmatrix}2\\-3\\2\end{pmatrix}-\begin{pmatrix}2\\5\\-1\end{pmatrix}=\begin{pmatrix}0\\-8\\3\end{pmatrix}$$

$x=2, \frac{y-5}{-8}=\frac{z+1}{3}$

7) e)

$$\begin{pmatrix}9\\8\\-6\end{pmatrix}-\begin{pmatrix}3\\8\\1\end{pmatrix}=\begin{pmatrix}6\\0\\-7\end{pmatrix}$$

$\frac{x-3}{6}=\frac{z-1}{-7}, y=8$

7) f)

$$\begin{pmatrix}6\\-7\\-2\end{pmatrix}-\begin{pmatrix}5\\-3\\-2\end{pmatrix}=\begin{pmatrix}1\\-4\\0\end{pmatrix}$$

$\frac{x-5}{1}=\frac{y+3}{-4}, z=-2$ or $x-5=\frac{y+3}{-4}, z=-2$

8)

Rewrite as:

$l_1: \frac{x-\frac{3}{2}}{3}=\frac{y+\frac{11}{5}}{-\frac{4}{5}}=\frac{z-\frac{9}{8}}{\frac{3}{2}}$

$l_2: \frac{3x-1}{-9}=\frac{10y-13}{8}=\frac{2z-5}{-3} \Rightarrow \frac{x-\frac{1}{3}}{-3}=\frac{y-\frac{13}{10}}{\frac{4}{5}}=\frac{z-\frac{5}{2}}{-\frac{3}{2}}$

The lines are parallel if there exists a k such that $\begin{pmatrix}3\\-\frac{4}{5}\\\frac{3}{2}\end{pmatrix}=k\begin{pmatrix}-3\\\frac{4}{5}\\-\frac{3}{2}\end{pmatrix}$

$3=-3k \Rightarrow k=-1$

$-\frac{4}{4}=\frac{4}{5}k \Rightarrow k=-1$

$\frac{3}{2}=-\frac{3}{2}k \Rightarrow k=-1$

As the values of k are consistent, the lines are parallel.

9)

$l_1: \mathbf{r} = \begin{pmatrix} 3 \\ -2 \\ 74 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ 3 \\ 10 \end{pmatrix}$

$l_2: \mathbf{r} = \begin{pmatrix} -1 \\ 4 \\ -2 \end{pmatrix} + \mu \begin{pmatrix} 3 \\ 6 \\ 8 \end{pmatrix}$

$3 + 2\lambda = -1 + 3\mu \Rightarrow 2\lambda - 3\mu = -4$ (1)

$-2 + 3\lambda = 4 + 6\mu \Rightarrow 3\lambda - 6\mu = 6$ (2)

$74 + 10\lambda = -2 + 8\mu \Rightarrow 10\lambda - 8\mu = -76$ (3)

Solve (1) and (2): $\lambda = -14$ and $\mu = -8$

Check in (3):

$10(-14) - 8(-8) = -76$ as required

So the lines intersect.

Substitute $\lambda = -14$ into l_1:

The lines intersect at $(-25, -44, -66)$

10)

Rewrite l_1 as $\frac{x+5}{-\frac{8}{5}} = \frac{y-5}{\frac{2}{5}} = \frac{z+\frac{3}{5}}{-\frac{4}{5}}$

Rewrite l_2 as $\frac{x+4}{-\frac{3}{5}} = \frac{y+5}{-\frac{3}{5}} = \frac{z-9}{\frac{2}{5}}$

$$l_1: \mathbf{r} = \begin{pmatrix} -5 \\ 5 \\ -\frac{3}{5} \end{pmatrix} + \lambda \begin{pmatrix} -\frac{8}{5} \\ \frac{2}{5} \\ -\frac{4}{5} \end{pmatrix} \text{ or } l_1: \mathbf{r} = \begin{pmatrix} -5 \\ 5 \\ -\frac{3}{5} \end{pmatrix} + \lambda \begin{pmatrix} -4 \\ 1 \\ -2 \end{pmatrix}$$

$$l_2: \mathbf{r} = \begin{pmatrix} -4 \\ -5 \\ 9 \end{pmatrix} + \mu \begin{pmatrix} -\frac{3}{5} \\ -\frac{3}{5} \\ \frac{2}{5} \end{pmatrix} \text{ or } l_2: \mathbf{r} = \begin{pmatrix} -4 \\ -5 \\ 9 \end{pmatrix} + \mu \begin{pmatrix} -3 \\ -3 \\ 2 \end{pmatrix}$$

$-5 - 4\lambda = -4 - 3\mu \Rightarrow -4\lambda + 3\mu = 1$ (1)

$5 + \lambda = -5 - 3\mu \Rightarrow \lambda + 3\mu = -10$ (2)

$-\frac{3}{5} - 2\lambda = 9 + 2\mu \Rightarrow -2\lambda - 2\mu = \frac{48}{5}$ (3)

Solve (1) and (2): $\lambda = -\frac{11}{5}$ and $\mu = -\frac{13}{5}$

Check in (3):

$-2\left(-\frac{11}{5}\right) - 2\left(-\frac{13}{5}\right) = \frac{48}{5}$ as required

So the lines intersect.

Substitute $\lambda = -\frac{11}{5}$ into l_1:

The lines intersect at $\left(\frac{19}{5}, \frac{14}{5}, \frac{19}{5}\right)$

11) a)

Rewrite as $l_1: \frac{x+1}{4} = \frac{y-3}{-5} = \frac{z-\frac{7}{2}}{-2}$

Rewrite as $l_2: \frac{x-9}{-4} = \frac{y+2}{5} = \frac{z+\frac{1}{3}}{2}$

$$l_1: \mathbf{r} = \begin{pmatrix} -1 \\ 3 \\ \frac{7}{2} \end{pmatrix} + \lambda \begin{pmatrix} 4 \\ -5 \\ -2 \end{pmatrix}$$

$$l_2: \mathbf{r} = \begin{pmatrix} 9 \\ -2 \\ -\frac{1}{3} \end{pmatrix} + \lambda \begin{pmatrix} -4 \\ 5 \\ 2 \end{pmatrix}$$

$4 = k(-4) \Rightarrow k = -1$

$-5 = k(5) \Rightarrow k = -1$

$-2 = k(2) \Rightarrow k = -1$

Consistent values of k, so the lines are parallel (or coincident).

Check if $\left(9, -2, -\frac{1}{3}\right)$ lies on l_1:

$9 = -1 + 4\lambda \Rightarrow \lambda = \frac{5}{2}$

$-2 = 3 - 5\lambda \Rightarrow \lambda = 1$

$-\frac{1}{3} = \frac{7}{2} - 2\lambda \Rightarrow \lambda = \frac{23}{12}$

Inconsistent values of λ, so the lines are not coincident

So the lines are parallel.

11) b)

Rewrite as l_1: $\frac{x+2}{3} = \frac{y-\frac{1}{4}}{\frac{3}{4}} = \frac{z-2}{-1}$

Rewrite as l_2: $\frac{x+3}{3} = \frac{y+\frac{3}{8}}{\frac{3}{4}} = \frac{z+3}{1}$

l_1: $\mathbf{r} = \begin{pmatrix} -2 \\ \frac{1}{4} \\ 2 \end{pmatrix} + \lambda \begin{pmatrix} 3 \\ \frac{3}{4} \\ -1 \end{pmatrix}$ or l_1: $\mathbf{r} = \begin{pmatrix} -2 \\ \frac{1}{4} \\ 2 \end{pmatrix} + \lambda \begin{pmatrix} 12 \\ 3 \\ -4 \end{pmatrix}$

l_2: $\mathbf{r} = \begin{pmatrix} -3 \\ -\frac{3}{8} \\ -3 \end{pmatrix} + \mu \begin{pmatrix} 3 \\ \frac{3}{4} \\ 1 \end{pmatrix}$ or l_2: $\mathbf{r} = \begin{pmatrix} -3 \\ -\frac{3}{8} \\ -3 \end{pmatrix} + \mu \begin{pmatrix} 12 \\ 3 \\ 4 \end{pmatrix}$

$12 = k(12) \Rightarrow k = 1$

$3 = k(3) \Rightarrow k = 1$

$-4 = k(4) \Rightarrow k = -1$

Inconsistent values of k, so the lines are not parallel (or coincident).

$-2 + 12\lambda = -3 + 12\mu \Rightarrow 12\lambda - 12\mu = -1$ (1)

$\frac{1}{4} + 3\lambda = -\frac{3}{8} + 3\mu \Rightarrow 3\lambda - 3\mu = -\frac{5}{8}$ (2)

$2 - 4\lambda = -3 + 4\mu \Rightarrow -4\lambda - 4\mu = -5$ (3)

Solve (2) and (3): $\lambda = \frac{25}{48}$ and $\mu = \frac{35}{48}$

Check in (1):

$12\left(\frac{25}{48}\right) - 12\left(\frac{35}{48}\right) = -\frac{5}{2} \neq -1$

So the lines do not intersect.

l_1 and l_2 are skew lines

11) c)

Rewrite as $l_1: \frac{x-\frac{1}{4}}{\frac{1}{2}} = \frac{y+7}{1} = \frac{z-3}{-8}$

Rewrite as $l_2: \frac{x-\frac{7}{6}}{\frac{1}{2}} = \frac{y+\frac{1}{2}}{2} = \frac{z-\frac{3}{2}}{1}$

$l_1: \mathbf{r} = \begin{pmatrix} \frac{1}{4} \\ -7 \\ 3 \end{pmatrix} + \lambda \begin{pmatrix} \frac{1}{2} \\ 1 \\ -8 \end{pmatrix}$ or $l_1: \mathbf{r} = \begin{pmatrix} \frac{1}{4} \\ -7 \\ 3 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ 2 \\ -16 \end{pmatrix}$

$l_2: \mathbf{r} = \begin{pmatrix} \frac{7}{6} \\ -\frac{1}{2} \\ \frac{3}{2} \end{pmatrix} + \mu \begin{pmatrix} \frac{1}{2} \\ 2 \\ 1 \end{pmatrix}$ or $l_2: \mathbf{r} = \begin{pmatrix} \frac{7}{6} \\ -\frac{1}{2} \\ \frac{3}{2} \end{pmatrix} + \mu \begin{pmatrix} 1 \\ 4 \\ 2 \end{pmatrix}$

$1 = k(1) \Rightarrow k = 1$

$2 = k(4) \Rightarrow k = \frac{1}{2}$

$-16 = k(2) \Rightarrow k = -8$

Inconsistent values of k, so the lines are not parallel (or coincident).

$\frac{1}{4} + \lambda = \frac{7}{6} + \mu \Rightarrow \lambda - \mu = \frac{11}{12}$ (1)

$-7 + 2\lambda = -\frac{1}{2} + 4\mu \Rightarrow 2\lambda - 4\mu = \frac{13}{2}$ (2)

$3 - 16\lambda = \frac{3}{2} + 2\mu \Rightarrow -16\lambda - 2\mu = -\frac{3}{2}$ (3)

Solve (1) and (2): $\lambda = -\frac{17}{12}$ and $\mu = -\frac{7}{3}$

Check in (3):

$-16\left(-\frac{17}{12}\right) - 2\left(-\frac{7}{3}\right) = \frac{82}{3} \neq -\frac{3}{2}$

So the lines do not intersect.

l_1 and l_2 are skew lines

11) d)

Rewrite as $l_1: \frac{4x-1}{-8} = \frac{y+5}{3} = \frac{5z-65}{4}$ and then as $l_1: \frac{x-\frac{1}{4}}{-2} = \frac{y+5}{3} = \frac{z-13}{\frac{4}{5}}$

Rewrite as $l_2: \frac{3x-2}{5} = \frac{4y-7}{-9} = \frac{z-3}{-1}$ and then as $l_2: \frac{x-\frac{2}{3}}{\frac{5}{3}} = \frac{y-\frac{7}{4}}{-\frac{9}{4}} = \frac{z-3}{-1}$

$l_1: \mathbf{r} = \begin{pmatrix} \frac{1}{4} \\ -5 \\ 13 \end{pmatrix} + \lambda \begin{pmatrix} -2 \\ 3 \\ \frac{4}{5} \end{pmatrix}$ or $l_1: \mathbf{r} = \begin{pmatrix} \frac{1}{4} \\ -5 \\ 13 \end{pmatrix} + \lambda \begin{pmatrix} -10 \\ 15 \\ 4 \end{pmatrix}$

$l_2: \mathbf{r} = \begin{pmatrix} \frac{2}{3} \\ \frac{7}{4} \\ 3 \end{pmatrix} + \mu \begin{pmatrix} \frac{5}{3} \\ -\frac{9}{4} \\ -1 \end{pmatrix}$ or $l_2: \mathbf{r} = \begin{pmatrix} \frac{2}{3} \\ \frac{7}{4} \\ 3 \end{pmatrix} + \mu \begin{pmatrix} 20 \\ -27 \\ -12 \end{pmatrix}$

$-10 = k(20) \Rightarrow k = -\frac{1}{2}$

$15 = k(-27) \Rightarrow k = -\frac{5}{9}$

$4 = k(-12) \Rightarrow k = -\frac{1}{3}$

Inconsistent values of k, so the lines are not parallel (or coincident).

$\frac{1}{4} - 10\lambda = \frac{2}{3} + 20\mu \Rightarrow -10\lambda - 20\mu = \frac{5}{12}$ (1)

$-5 + 15\lambda = \frac{7}{4} - 27\mu \Rightarrow 15\lambda + 27\mu = \frac{27}{4}$ (2)

$13 + 4\lambda = 3 - 12\mu \Rightarrow 4\lambda + 12\mu = -10$ (3)

Solve (1) and (2): $\lambda = \frac{39}{8}$ and $\mu = -\frac{59}{24}$

Check in (3):

$4\left(\frac{39}{8}\right) + 12\left(-\frac{59}{24}\right) = -10$ as required

So the lines do intersect.

Substitute $\lambda = \frac{39}{8}$ into l_1 to find where the lines intersect:

$\left(\frac{1}{4} + \left(\frac{39}{8}\right) \times -10, -5 + \left(\frac{39}{8}\right) \times 15, 13 + \left(\frac{39}{8}\right) \times 4\right) = \left(-\frac{97}{2}, \frac{545}{8}, \frac{65}{2}\right)$

12) a)

$l_1: \mathbf{r} = \begin{pmatrix} 6 \\ -1 \\ 7 \end{pmatrix} + \lambda \begin{pmatrix} -3 \\ 2 \\ 4 \end{pmatrix}$

$l_2: \mathbf{r} = \begin{pmatrix} 15 \\ -3 \\ -12 \end{pmatrix} + \mu \begin{pmatrix} k \\ 9 \\ 6 \end{pmatrix}$

$6 - 3\lambda = 15 + k\mu \Rightarrow -3\lambda - k\mu = 9$ (1)

$-1 + 2\lambda = -3 + 9\mu \Rightarrow 2\lambda - 9\mu = -2$ (2)

$7 + 4\lambda = -12 + 6\mu \Rightarrow 4\lambda - 6\mu = -19$ (3)

Solve (2) and (3): $\lambda = -\frac{53}{8}$ and $\mu = -\frac{5}{4}$

Substitute into (1):

$-3\left(-\frac{53}{8}\right) - k\left(-\frac{5}{4}\right) = 9$

$\Rightarrow \frac{159}{8} + \frac{5}{4}k = 9$

$\Rightarrow k = -\frac{87}{10}$

12) b)

Substitute $\lambda = -\frac{53}{8}$ into l_1 to find where the lines intersect:

$\left(6 + \left(-\frac{53}{8}\right) \times (-3), -1 + \left(-\frac{53}{8}\right) \times 2, 7 + \left(-\frac{53}{8}\right) \times 4\right) = \left(\frac{207}{8}, -\frac{57}{4}, -\frac{39}{2}\right)$

The position vector of the point of intersection is $\begin{pmatrix} \frac{207}{8} \\ -\frac{57}{4} \\ -\frac{39}{2} \end{pmatrix}$

13) a)

$l_1: \mathbf{r} = \begin{pmatrix} 3 \\ -7 \\ 3 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ 9 \\ 4 \end{pmatrix}$

$l_2: \mathbf{r} = \begin{pmatrix} -7 \\ 1 \\ -8 \end{pmatrix} + \mu \begin{pmatrix} 4 \\ 2 \\ 3 \end{pmatrix}$

$\cos\theta = \frac{\begin{pmatrix} 2 \\ 9 \\ 4 \end{pmatrix} \cdot \begin{pmatrix} 4 \\ 2 \\ 3 \end{pmatrix}}{\left\| \begin{pmatrix} 2 \\ 9 \\ 4 \end{pmatrix} \right\| \left\| \begin{pmatrix} 4 \\ 2 \\ 3 \end{pmatrix} \right\|} = \frac{2\times4+9\times2+4\times3}{\sqrt{2^2+9^2+4^2}\sqrt{4^2+2^2+3^2}} = \frac{38}{\sqrt{101}\sqrt{29}} = \frac{38}{\sqrt{2929}}$

$\theta = \arccos\left(\frac{38}{\sqrt{2929}}\right) = 45.4°$ (1 DP)

13) b)

Rewrite as $l_1: \frac{x+2}{-8} = \frac{y-1}{2} = z+2$ and then as $l_1: \frac{x+2}{-8} = \frac{y-1}{2} = \frac{z+2}{1}$

$l_1: \mathbf{r} = \begin{pmatrix} -2 \\ 1 \\ -2 \end{pmatrix} + \lambda \begin{pmatrix} -8 \\ 2 \\ 1 \end{pmatrix}$

$l_2: \mathbf{r} = \begin{pmatrix} 3 \\ -2 \\ 4 \end{pmatrix} + \mu \begin{pmatrix} 4 \\ 3 \\ 2 \end{pmatrix}$

$\cos\theta = \frac{\begin{pmatrix} -8 \\ 2 \\ 1 \end{pmatrix} \cdot \begin{pmatrix} 4 \\ 3 \\ 2 \end{pmatrix}}{\left|\begin{pmatrix} -8 \\ 2 \\ 1 \end{pmatrix}\right| \left|\begin{pmatrix} 4 \\ 3 \\ 2 \end{pmatrix}\right|} = \frac{(-8)\times 4 + 2\times 3 + 1\times 2}{\sqrt{(-8)^2+2^2+1^2}\sqrt{4^2+3^2+2^2}} = \frac{-24}{\sqrt{69}\sqrt{29}} = -\frac{8\sqrt{2001}}{667}$

$\theta = \arccos\left(-\frac{8\sqrt{2001}}{667}\right) = 122.447\ldots°$

Acute Angle $= 180° - 122.447\ldots° = 57.6°$ (1 DP)

13) c)

Rewrite as $l_1: \frac{3-x}{2} = \frac{y+1}{-3} = 3-z$ and then as $l_1: \frac{x-3}{-2} = \frac{y+1}{-3} = \frac{z-3}{-1}$

Rewrite as $l_2: x+2 = 3-y = 4-z$ and then as $l_2: \frac{x+2}{1} = \frac{y-3}{-1} = \frac{z-4}{-1}$

$l_1: \mathbf{r} = \begin{pmatrix} 3 \\ -1 \\ 3 \end{pmatrix} + \lambda \begin{pmatrix} -2 \\ -3 \\ -1 \end{pmatrix}$ or $l_1: \mathbf{r} = \begin{pmatrix} 3 \\ -1 \\ 3 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ 3 \\ 1 \end{pmatrix}$

$l_2: \mathbf{r} = \begin{pmatrix} -2 \\ 3 \\ 4 \end{pmatrix} + \mu \begin{pmatrix} 1 \\ -1 \\ -1 \end{pmatrix}$

$\cos\theta = \frac{\begin{pmatrix} 2 \\ 3 \\ 1 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ -1 \\ -1 \end{pmatrix}}{\left|\begin{pmatrix} 2 \\ 3 \\ 1 \end{pmatrix}\right| \left|\begin{pmatrix} 1 \\ -1 \\ -1 \end{pmatrix}\right|} = \frac{2\times 1 + 3\times(-1) + 1\times(-1)}{\sqrt{2^2+3^2+1^2}\sqrt{1^2+(-1)^2+(-1)^2}} = \frac{-2}{\sqrt{14}\sqrt{3}} = -\frac{\sqrt{42}}{21}$

$\theta = \arccos\left(-\frac{\sqrt{42}}{21}\right) = 107.975\ldots°$

Acute Angle $= 180° - 107.975\ldots° = 72.0°$ (1 DP)

13) d)

Rewrite as $l_1: x = 3, \frac{2-y}{2} = z + 8$ and then as $l_1: x = 3, \frac{y-2}{-2} = \frac{z+8}{1}$

Rewrite as $l_2: x = 8, y - 9 = \frac{z-4}{5}$ and then as $l_2: x = 8, \frac{y-9}{1} = \frac{z-4}{5}$

$$l_1: \mathbf{r} = \begin{pmatrix} 3 \\ 2 \\ -8 \end{pmatrix} + \lambda \begin{pmatrix} 0 \\ -2 \\ 1 \end{pmatrix}$$

$$l_2: \mathbf{r} = \begin{pmatrix} 8 \\ 9 \\ 4 \end{pmatrix} + \mu \begin{pmatrix} 0 \\ 1 \\ 5 \end{pmatrix}$$

$$\cos\theta = \frac{\begin{pmatrix} 0 \\ -2 \\ 1 \end{pmatrix} \cdot \begin{pmatrix} 0 \\ 1 \\ 5 \end{pmatrix}}{\left|\begin{pmatrix} 0 \\ -2 \\ 1 \end{pmatrix}\right| \left|\begin{pmatrix} 0 \\ 1 \\ 5 \end{pmatrix}\right|} = \frac{0\times0+(-2)\times1+1\times5}{\sqrt{0^2+(-2)^2+1^2}\sqrt{0^2+1^2+5^2}} = \frac{3}{\sqrt{5}\sqrt{26}} = \frac{3\sqrt{130}}{130}$$

$$\theta = \arccos\left(\frac{3\sqrt{130}}{130}\right) = 74.7° \text{ (1 DP)}$$

13) e)

Rewrite as $l_1: \frac{2x-3}{4} = \frac{3y-7}{-6} = \frac{2z+1}{4}$ and then as $l_1: \frac{x-\frac{3}{2}}{2} = \frac{y-\frac{7}{3}}{-2} = \frac{z+\frac{1}{2}}{2}$

Rewrite as $l_2: \frac{2x+1}{3} = \frac{4y-6}{-5} = \frac{z-10}{-1}$ and then as $l_2: \frac{x+\frac{1}{2}}{\frac{3}{2}} = \frac{y-\frac{3}{2}}{-\frac{5}{4}} = \frac{z-10}{-1}$

$$l_1: \mathbf{r} = \begin{pmatrix} \frac{3}{2} \\ \frac{7}{3} \\ -\frac{1}{2} \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ -2 \\ 2 \end{pmatrix} \text{ or } l_1: \mathbf{r} = \begin{pmatrix} \frac{3}{2} \\ \frac{7}{3} \\ -\frac{1}{2} \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix}$$

$$l_2: \mathbf{r} = \begin{pmatrix} -\frac{1}{2} \\ \frac{3}{2} \\ 10 \end{pmatrix} + \mu \begin{pmatrix} \frac{3}{2} \\ -\frac{5}{4} \\ -1 \end{pmatrix} \text{ or } l_2: \mathbf{r} = \begin{pmatrix} -\frac{1}{2} \\ \frac{3}{2} \\ 10 \end{pmatrix} + \mu \begin{pmatrix} 6 \\ -5 \\ -4 \end{pmatrix}$$

$$\cos\theta = \frac{\begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix} \cdot \begin{pmatrix} 6 \\ -5 \\ -4 \end{pmatrix}}{\left|\begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix}\right| \left|\begin{pmatrix} 6 \\ -5 \\ -4 \end{pmatrix}\right|} = \frac{1\times6+(-1)\times(-5)+1\times(-4)}{\sqrt{1^2+(-1)^2+1^2}\sqrt{6^2+(-5)^2+(-4)^2}} = \frac{7}{\sqrt{3}\sqrt{77}} = \frac{\sqrt{231}}{33}$$

$$\theta = \arccos\left(\frac{\sqrt{231}}{33}\right) = 62.6° \text{ (1 DP)}$$

13) f)

Rewrite as $l_1: \frac{x-5}{-1} = \frac{y-9}{4}, z = 5$

Rewrite as $l_2: \frac{8x+3}{3} = \frac{z-4}{5}, y = -2$ and then as $l_2: \frac{x+\frac{3}{8}}{\frac{3}{8}} = \frac{z-4}{5}, y = -2$

$l_1: \mathbf{r} = \begin{pmatrix} 5 \\ 9 \\ 5 \end{pmatrix} + \lambda \begin{pmatrix} -1 \\ 4 \\ 0 \end{pmatrix}$

$l_2: \mathbf{r} = \begin{pmatrix} -\frac{3}{8} \\ -2 \\ 4 \end{pmatrix} + \mu \begin{pmatrix} \frac{3}{8} \\ 0 \\ 5 \end{pmatrix}$ or $l_2: \mathbf{r} = \begin{pmatrix} -\frac{3}{8} \\ -2 \\ 4 \end{pmatrix} + \mu \begin{pmatrix} 3 \\ 0 \\ 40 \end{pmatrix}$

$\cos\theta = \frac{\begin{pmatrix} -1 \\ 4 \\ 0 \end{pmatrix} \cdot \begin{pmatrix} 3 \\ 0 \\ 40 \end{pmatrix}}{\left|\begin{pmatrix} -1 \\ 4 \\ 0 \end{pmatrix}\right| \left|\begin{pmatrix} 3 \\ 0 \\ 40 \end{pmatrix}\right|} = \frac{(-1)\times 3 + 4\times 0 + 0\times 40}{\sqrt{(-1)^2+4^2+0^2}\sqrt{3^2+0^2+40^2}} = \frac{-3}{\sqrt{17}\sqrt{1609}} = -\frac{3}{\sqrt{27353}}$

$\theta = \arccos\left(-\frac{3}{\sqrt{27353}}\right) = 91.039\ldots°$

Acute angle is $180° - 91.039\ldots° = 89.0°$ (1 DP)

14) a)

Rewrite as $l_2: \frac{x+200}{15k} = \frac{y-380}{18} = \frac{z-475}{-5}$

$l_1: \mathbf{r} = \begin{pmatrix} 2140 \\ 425 \\ 670 \end{pmatrix} + \lambda \begin{pmatrix} 10 \\ 9 \\ 0 \end{pmatrix}$

$l_2: \mathbf{r} = \begin{pmatrix} -200 \\ 380 \\ 475 \end{pmatrix} + \mu \begin{pmatrix} 15k \\ 18 \\ -5 \end{pmatrix}$

$2140 + 10\lambda = -200 + 15k\mu \Rightarrow 10\lambda - 15k\mu = -2340$ (1)

$425 + 9\lambda = 380 + 18\mu \Rightarrow 9\lambda - 18\mu = -45$ (2)

$670 = 475 - 5\mu \Rightarrow \mu = -39$ (3)

Solve (2) and (3): $\lambda = -83$ and $\mu = -39$

Substitute into (1):

$10(-83) - 15k(-39) = -2340$

$\Rightarrow -830 + 585k = -2340$

$\Rightarrow k = -\frac{302}{117}$

14) b)

Substitute $\lambda = -83$ into l_1 to find where the lines intersect:

$(2140 + (-83) \times 10,425 + (-83) \times 9,670) = (1310, -322, 670)$

14) c)

The asteroids may be travelling at different speeds.

15) a)

Rewrite as $l_1: \frac{x+1}{4} = \frac{y-3}{-1} = \frac{3z+4}{6}$ and then as $l_1: \frac{x+1}{4} = \frac{y-3}{-1} = \frac{z+\frac{4}{3}}{2}$

$$l_1: \mathbf{r} = \begin{pmatrix} -1 \\ 3 \\ -\frac{4}{3} \end{pmatrix} + \lambda \begin{pmatrix} 4 \\ -1 \\ 2 \end{pmatrix}$$

$$\begin{pmatrix} 6 \\ k \\ k+1 \end{pmatrix} - \begin{pmatrix} 5 \\ 0 \\ 4 \end{pmatrix} = \begin{pmatrix} 1 \\ k \\ k-3 \end{pmatrix}$$

$$l_2: \mathbf{r} = \begin{pmatrix} 5 \\ 0 \\ 4 \end{pmatrix} + \mu \begin{pmatrix} 1 \\ k \\ k-3 \end{pmatrix}$$

l_1 and l_2 are perpendicular so:

$$\begin{pmatrix} 4 \\ -1 \\ 2 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ k \\ k-3 \end{pmatrix} = 0$$

$\Rightarrow 4 \times 1 + (-1) \times k + 2 \times (k-3) = 0$

$\Rightarrow 4 - k + 2k - 6 = 0$

$\Rightarrow k - 2 = 0 \Rightarrow k = 2$

15) b)

$$\begin{pmatrix} 6 \\ k \\ k-1 \end{pmatrix} - \begin{pmatrix} 5 \\ 0 \\ 4 \end{pmatrix} = \begin{pmatrix} 1 \\ k \\ k-5 \end{pmatrix}$$

As $k = 2$, the direction vector of l_3 is $\begin{pmatrix} 1 \\ 2 \\ -3 \end{pmatrix}$

$$l_3: \mathbf{r} = \begin{pmatrix} 5 \\ 0 \\ 4 \end{pmatrix} + \gamma \begin{pmatrix} 1 \\ 2 \\ -3 \end{pmatrix}$$

$$\cos\theta = \frac{\begin{pmatrix} 4 \\ -1 \\ 2 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ 2 \\ -3 \end{pmatrix}}{\left|\begin{pmatrix} 4 \\ -1 \\ 2 \end{pmatrix}\right| \left|\begin{pmatrix} 1 \\ 2 \\ -3 \end{pmatrix}\right|} = \frac{4\times1+(-1)\times2+2\times(-3)}{\sqrt{4^2+(-1)^2+2^2}\sqrt{1^2+2^2+(-3)^2}}$$

$$= \frac{-4}{\sqrt{21}\sqrt{14}} = -\frac{4}{\sqrt{294}} = -\frac{4}{7\sqrt{6}} \times \frac{\sqrt{6}}{\sqrt{6}} = -\frac{4\sqrt{6}}{42} = -\frac{2\sqrt{6}}{21}$$

16) a)

Rewrite as $l_1: \frac{x-5}{-2} = \frac{y-3}{-1} = \frac{2z}{1}$ and then as $l_1: \frac{x-5}{-2} = \frac{y-3}{-1} = \frac{z}{\frac{1}{2}}$

Rewrite as $l_2: \frac{x-2}{8-k} = \frac{y+5}{2p+5} = \frac{z-6}{-4}$

$l_1: \mathbf{r} = \begin{pmatrix} 5 \\ 3 \\ 0 \end{pmatrix} + \lambda \begin{pmatrix} -2 \\ -1 \\ \frac{1}{2} \end{pmatrix}$ or $l_1: \mathbf{r} = \begin{pmatrix} 5 \\ 3 \\ 0 \end{pmatrix} + \lambda \begin{pmatrix} -4 \\ -2 \\ 1 \end{pmatrix}$

$l_2: \mathbf{r} = \begin{pmatrix} 2 \\ -5 \\ 6 \end{pmatrix} + \mu \begin{pmatrix} 8-k \\ 2p+5 \\ -4 \end{pmatrix}$

l_1 and l_2 are parallel:

$\begin{pmatrix} -4 \\ -2 \\ 1 \end{pmatrix} = q \begin{pmatrix} 8-k \\ 2p+5 \\ -4 \end{pmatrix}$

$-4 = q(8-k)$ (1)

$-2 = q(2p+5)$ (2)

$1 = q(-4)$ (3)

Solve (3) to get $q = -\frac{1}{4}$

Substitute into (1):

$-4 = -\frac{1}{4}(8-k) \Rightarrow k = -8$

Substitute into (2)

$-2 = -\frac{1}{4}(2p+5) \Rightarrow p = \frac{3}{2}$

So $k = -8$ and $p = \frac{3}{2}$

16) b)

$l_1: \mathbf{r} = \begin{pmatrix} 5 \\ 3 \\ 0 \end{pmatrix} + \lambda \begin{pmatrix} -4 \\ -2 \\ 1 \end{pmatrix}$

$l_2: \mathbf{r} = \begin{pmatrix} 2 \\ -5 \\ 6 \end{pmatrix} + \mu \begin{pmatrix} 8-k \\ 2p+5 \\ -4 \end{pmatrix}$

l_1 and l_2 intersect at right angles:

$\begin{pmatrix} -4 \\ -2 \\ 1 \end{pmatrix} \cdot \begin{pmatrix} 8-k \\ 2p+5 \\ -4 \end{pmatrix} = 0$

$\Rightarrow (-4) \times (8-k) + (-2) \times (2p+5) + 1 \times (-4) = 0$

$\Rightarrow -32 + 4k - 4p - 10 - 4 = 0$

$\Rightarrow 4k - 4p = 46$

$\Rightarrow k = p + \frac{23}{2}$ (1)

Now find where they intersect:

$5 - 4\lambda = 2 + (8-k)\mu$ (2)

$3 - 2\lambda = -5 + (2p+5)\mu$ (3)

$\lambda = 6 - 4\mu$ (4)

Substitute (1) & (4) into (2):

$5 - 4(6 - 4\mu) = 2 + \left(8 - \left(p + \frac{23}{2}\right)\right)\mu$

$\Rightarrow -19 + 16\mu = 2 - p\mu - \frac{7}{2}\mu$

$\Rightarrow p\mu = -\frac{39}{2}\mu + 21$

$\Rightarrow p = -\frac{39}{2} + \frac{21}{\mu}$ (5)

Substitute (4) into (3):

$3 - 2(6 - 4\mu) = -5 + (2p+5)\mu$

$\Rightarrow -9 + 8\mu = -5 + 2p\mu + 5\mu$

$\Rightarrow 2p\mu = -4 + 3\mu$

$\Rightarrow p = -\frac{2}{\mu} + \frac{3}{2}$ (6)

Equating (5) & (6):

$-\frac{39}{2} + \frac{21}{\mu} = -\frac{2}{\mu} + \frac{3}{2}$

$\Rightarrow -39\mu + 42 = -4 + 3\mu$

$\Rightarrow 42\mu - 46 = 0$

$\Rightarrow \mu = \frac{23}{21}$

Then $p = -\frac{2}{\left(\frac{23}{21}\right)} + \frac{3}{2} = -\frac{15}{46}$

And $k = \left(-\frac{15}{46}\right) + \frac{23}{2} = \frac{257}{23}$

17)

Rewrite as $l_1: \frac{x-3}{7} = \frac{2y+1}{2} = \frac{z-1}{-1}$ and then as $l_1: \frac{x-3}{7} = \frac{y+\frac{1}{2}}{1} = \frac{z-1}{-1}$

Rewrite as $l_2: \frac{x-8}{3} = \frac{y-\frac{21}{2}}{3} = \frac{z+4}{-\frac{3}{2}}$

$$l_1: \mathbf{r} = \begin{pmatrix} 3 \\ -\frac{1}{2} \\ 1 \end{pmatrix} + \lambda \begin{pmatrix} 7 \\ 1 \\ -1 \end{pmatrix}$$

$$l_2: \mathbf{r} = \begin{pmatrix} 8 \\ \frac{21}{2} \\ -4 \end{pmatrix} + \mu \begin{pmatrix} 3 \\ 3 \\ -\frac{3}{2} \end{pmatrix} \text{ or } l_2: \mathbf{r} = \begin{pmatrix} 8 \\ \frac{21}{2} \\ -4 \end{pmatrix} + \mu \begin{pmatrix} 2 \\ 2 \\ -1 \end{pmatrix}$$

$$\Pi: \mathbf{r} = \begin{pmatrix} 3 \\ -\frac{1}{2} \\ 1 \end{pmatrix} + \lambda \begin{pmatrix} 7 \\ 1 \\ -1 \end{pmatrix} + \mu \begin{pmatrix} 2 \\ 2 \\ -1 \end{pmatrix}$$

18)

Rewrite as $l_1: \frac{x-3}{-1} = \frac{y-4}{5} = \frac{z+1}{1}$

Rewrite as $l_2: \frac{x+2}{2} = \frac{y-2}{-1} = \frac{z+20}{6}$

$$l_1: \mathbf{r} = \begin{pmatrix} 3 \\ 4 \\ -1 \end{pmatrix} + \lambda \begin{pmatrix} -1 \\ 5 \\ 1 \end{pmatrix}$$

$$l_2: \mathbf{r} = \begin{pmatrix} -2 \\ 2 \\ -20 \end{pmatrix} + \mu \begin{pmatrix} 2 \\ -1 \\ 6 \end{pmatrix}$$

$$\begin{pmatrix} -1 \\ 5 \\ 1 \end{pmatrix} \times \begin{pmatrix} 2 \\ -1 \\ 6 \end{pmatrix} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ -1 & 5 & 1 \\ 2 & -1 & 6 \end{vmatrix}$$

$$= \big(5 \times 6 - 1 \times (-1)\big)\mathbf{i} - \big((-1) \times 6 - 1 \times 2\big)\mathbf{j} + \big((-1) \times (-1) - 5 \times 2\big)\mathbf{k}$$

$$= 31\mathbf{i} + 8\mathbf{j} - 9\mathbf{k}$$

So $\Pi: 31x + 8y - 9z = d$

Substitute $(3, 4, -1)$ into the equation to find d:

$31(3) + 8(4) - 9(-1) = d \Rightarrow d = 134$

$\therefore \Pi: 31x + 8y - 9z = 134$

19) a)

$$\mathbf{r} = \begin{pmatrix} -1 \\ 1 \\ -2 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ 3 \\ -3 \end{pmatrix}$$

19) b)

$$\mathbf{r} = \begin{pmatrix} 0 \\ -9 \\ 1 \end{pmatrix} + \lambda \begin{pmatrix} 8 \\ 4 \\ -5 \end{pmatrix}$$

19) c)

Rewrite as $\frac{x+8}{3} = \frac{y-2}{-3} = \frac{z}{10}$

$$\mathbf{r} = \begin{pmatrix} -8 \\ 2 \\ 0 \end{pmatrix} + \lambda \begin{pmatrix} 3 \\ -3 \\ 10 \end{pmatrix}$$

19) d)

Rewrite as $\frac{x-7}{-5} = \frac{y-8}{12} = \frac{z+3}{1}$

$$\mathbf{r} = \begin{pmatrix} 7 \\ 8 \\ -3 \end{pmatrix} + \lambda \begin{pmatrix} -5 \\ 12 \\ 1 \end{pmatrix}$$

19) e)

Rewrite as $\frac{x-\frac{3}{2}}{\frac{5}{2}} = \frac{y+\frac{1}{3}}{\frac{2}{3}} = \frac{z-\frac{3}{5}}{\frac{2}{5}}$

$$\mathbf{r} = \begin{pmatrix} \frac{3}{2} \\ -\frac{1}{3} \\ \frac{3}{5} \end{pmatrix} + \lambda \begin{pmatrix} \frac{5}{2} \\ \frac{2}{3} \\ \frac{2}{5} \end{pmatrix} \text{ or } \mathbf{r} = \begin{pmatrix} \frac{3}{2} \\ -\frac{1}{3} \\ \frac{3}{5} \end{pmatrix} + \lambda \begin{pmatrix} 75 \\ 20 \\ 12 \end{pmatrix}$$

19) f)

Rewrite as $\frac{4x-3}{-6}=\frac{2y-5}{-2}=\frac{8z-6}{5}$

And then as $\frac{x-\frac{3}{4}}{-\frac{3}{2}}=\frac{y-\frac{5}{2}}{-1}=\frac{z-\frac{3}{4}}{\frac{5}{8}}$

$$\mathbf{r}=\begin{pmatrix}\frac{3}{4}\\ \frac{5}{2}\\ \frac{3}{4}\end{pmatrix}+\lambda\begin{pmatrix}-\frac{3}{2}\\ -1\\ \frac{5}{8}\end{pmatrix} \text{ or } \mathbf{r}=\begin{pmatrix}\frac{3}{4}\\ \frac{5}{2}\\ \frac{3}{4}\end{pmatrix}+\lambda\begin{pmatrix}-12\\ -8\\ 5\end{pmatrix}$$

19) g)

$$\mathbf{r}=\begin{pmatrix}-4\\ 3\\ -12\end{pmatrix}+\lambda\begin{pmatrix}0\\ 4\\ 7\end{pmatrix}$$

19) h)

Rewrite as $\frac{x-9}{1}=\frac{y-2}{-\frac{5}{2}}, z=1$

$$\mathbf{r}=\begin{pmatrix}9\\ 2\\ 1\end{pmatrix}+\lambda\begin{pmatrix}1\\ -\frac{5}{2}\\ 0\end{pmatrix} \text{ or } \mathbf{r}=\begin{pmatrix}9\\ 2\\ 1\end{pmatrix}+\lambda\begin{pmatrix}2\\ -5\\ 0\end{pmatrix}$$

19) i)

Rewrite as $\frac{x}{1}=\frac{z-6}{-1}, y=5$

$$\mathbf{r}=\begin{pmatrix}0\\ 5\\ 6\end{pmatrix}+\lambda\begin{pmatrix}1\\ 0\\ -1\end{pmatrix}$$

20) a)

$$\left(\mathbf{r}-\begin{pmatrix}3\\ -1\\ 4\end{pmatrix}\right)\times\begin{pmatrix}5\\ 2\\ 6\end{pmatrix}=\mathbf{0}$$

20) b)

Rewrite as $\frac{x}{9} = \frac{y-2}{-3} = \frac{z-5}{-1}$

$$\left(\mathbf{r} - \begin{pmatrix} 0 \\ 2 \\ 5 \end{pmatrix}\right) \times \begin{pmatrix} 9 \\ -3 \\ -1 \end{pmatrix} = \mathbf{0}$$

20) c)

Rewrite as $\frac{x-3}{-1} = \frac{y+\frac{1}{2}}{1} = \frac{z-3}{-1}$

$$\left(\mathbf{r} - \begin{pmatrix} 3 \\ -\frac{1}{2} \\ 3 \end{pmatrix}\right) \times \begin{pmatrix} -1 \\ 1 \\ -1 \end{pmatrix} = \mathbf{0}$$

20) d)

Rewrite as $\frac{x+\frac{2}{3}}{\frac{5}{3}} = \frac{y-6}{-1}$, $z = 2$

$$\left(\mathbf{r} - \begin{pmatrix} -\frac{2}{3} \\ 6 \\ 2 \end{pmatrix}\right) \times \begin{pmatrix} \frac{5}{3} \\ -1 \\ 0 \end{pmatrix} = \mathbf{0}$$

20) e)

Rewrite as $x = \frac{1}{3}$, $\frac{y-\frac{3}{4}}{\frac{1}{2}} = \frac{z-\frac{5}{3}}{-\frac{5}{3}}$

$$\left(\mathbf{r} - \begin{pmatrix} \frac{1}{3} \\ \frac{3}{4} \\ \frac{5}{3} \end{pmatrix}\right) \times \begin{pmatrix} 0 \\ \frac{1}{2} \\ -\frac{5}{3} \end{pmatrix} = \mathbf{0}$$

20) f)

Rewrite as $x = 4$, $\frac{y}{\frac{1}{3}} = \frac{z-2}{-\frac{1}{2}}$

$$\left(\mathbf{r} - \begin{pmatrix} 4 \\ 0 \\ 2 \end{pmatrix}\right) \times \begin{pmatrix} 0 \\ \frac{1}{3} \\ -\frac{1}{2} \end{pmatrix} = \mathbf{0}$$